

Jeremy McMahan

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Education

- 2018–2025 **PhD in Computer Science**, *University of Wisconsin-Madison*
- Thesis: Safe Multi-Agent Reinforcement Learning in Polynomial Time
 - Advisor: Prof. Jerry Zhu
 - Distinctions: LEAP Fellow
- 2018–2020 **MS in Computer Science**, *University of Wisconsin-Madison*
- Thesis: A D-competitive algorithm for the Multilevel Aggregation Problem with Deadlines
 - Advisor: Prof. Shuchi Chawla
- 2016–2018 **BS in Mathematics and Computer Science**, *University of Illinois Urbana-Champaign*
- Thesis: Spectral Graph Isomorphism
 - Advisor: Prof. Alexandra Kolla
 - Distinctions: Ranked #1 in Mathematics and Computer Science

Research Summary

Designing polynomial-time algorithms for decision-making in combinatorial and strategic settings. My research bridges **Combinatorial Optimization**, **Algorithmic Game Theory**, and **Reinforcement Learning** to develop approximation frameworks with provable performance guarantees for constrained and robust multi-agent sequential decision problems, yielding general algorithmic approaches applicable to routing, resource allocation, digital marketplaces, and safety-critical autonomous systems.

Research Experience

- 2022–2024 **IFDS Research Assistantship**, *Institute for Foundations of Data Science*
Selected for a competitive, grant-funded interdisciplinary research position supporting the Institute's mission to develop the theoretical foundations of Data Science.
- 2020–2025 **Graduate Research Assistant**, *University of Wisconsin-Madison*
Conducted doctoral research on the algorithmic foundations of decision-making under constraints and strategic interaction, resulting in publications at top venues including NeurIPS, ICML, and AAAI.

Constrained Sequential Decision-Making

Constrained MDPs: Designed the first polynomial-time approximation frameworks for constrained Markov Decision Processes, resolving long-standing open problems on the tractability of constrained decision-making. These works yield general algorithmic templates for improving the feasibility and reliability of constrained autonomous decision-making systems. (*ICML '25, NeurIPS '24, AISTATS '24*)

Online & Stochastic Optimization: Designed polynomial-time algorithms for online and adaptive decision-making under uncertainty, including state-of-the-art results for the Multilevel Aggregation Problem and the first adaptive approximation algorithms for correlated Pandora's Box. These works provide general approaches for real-time routing, resource allocation, and inventory management. (*APPROX '23, Preprint*)

Strategic Multi-Agent Decision-Making

Robust Equilibria: Developed the first polynomial-time algorithms for computing anytime-constrained, adversarial-defense, and uncertainty-robust equilibria for Markov Games, resolving fundamental tractability challenges in robust decision-making. These results provide unified algorithmic frameworks for enhancing the robustness and safety of multi-agent systems. (*ICML '25, ICML '24, AAAI '24*)

Adversarial Attacks: Developed optimal, polynomial-time algorithms for offline reward poisoning, online adversarial attacks, and offline misinformation attacks on Markov Games, identifying the fundamental vulnerability limits of strategic manipulation. This work yields general-purpose tools for stress-testing and hardening multi-agent systems. (*ICML '24, RLC '24, AAAI '24, AAAI '23*)

Publications

** Indicates equal contribution. Authorship order follows venue conventions.*

Refereed Conference Proceedings

- ICML '25 **Polynomial-Time Approximability of Constrained Reinforcement Learning**
Jeremy McMahan
- ICML '25 **Anytime-Constrained Equilibria in Polynomial Time**
Jeremy McMahan
- NeurIPS '24 **Deterministic Policies for Constrained Reinforcement Learning in Polynomial Time**
Jeremy McMahan
- RLC '24 **Inception: Efficiently Computable Misinformation Attacks on Markov Games**
Jeremy McMahan, Young Wu, Yudong Chen, Xiaojin Zhu, Qiaomin Xie
[Oral]

- ICML '24 **Roping in Uncertainty: Robustness and Regularization in Markov Games**
Jeremy McMahan, Giovanni Artiglio, Qiaomin Xie
- ICML '24 **Minimally Modifying a Markov Game to Achieve any Nash Equilibrium and Value**
 Young Wu, *Jeremy McMahan*, Yiding Chen, Yudong Chen, Xiaojin Zhu, Qiaomin Xie
- CogSci '24 **Various Misleading Visual Features in Misleading Graphs: Do they truly deceive us?**
 Jihyun Rho, Martina A Rau, Shubham Kumar Bharti, Rosanne Luu, *Jeremy McMahan*, Andrew Wang, Jerry Zhu
- AISTATS '24 **Anytime-Constrained Reinforcement Learning**
Jeremy McMahan, Xiaojin Zhu
- AAAI '24 **Optimal Attack and Defense for Reinforcement Learning**
Jeremy McMahan, Young Wu, Xiaojin Zhu, Qiaomin Xie
[\[Oral\]](#)
- AAAI '24 **Data Poisoning to Fake a Nash Equilibrium for Markov Games**
 Young Wu, *Jeremy McMahan*, Xiaojin Zhu, Qiaomin Xie
- APPROX '23 **Approximating Pandora's Box with Correlations**
 Shuchi Chawla*, Evangelia Gergatsouli*, *Jeremy McMahan**, Christos Tzamos*
[\[Oral\]](#)
- AAAI '23 **Reward Poisoning Attacks on Offline Multi-Agent Reinforcement Learning**
 Young Wu, *Jeremy McMahan*, Xiaojin Zhu, Qiaomin Xie
[\[Oral\]](#)
- GAMES '20 **Noble Deceit: Optimizing Social Welfare for Myopic Multi-Armed Bandits**
 Ashwin Maran*, *Jeremy McMahan**, Nathaniel Sauerberg*
[Preprints and Working Papers](#)
- Preprint **A D-competitive Algorithm for the Multilevel Aggregation Problem with Deadlines**
Jeremy McMahan

Software and Research Artifacts

- Python **Anytime-Constrained Decision-Making**, [\[GitHub Link\]](#)
 A reusable algorithmic framework for constrained decision-making, providing polynomial-time optimization solvers and Gymnasium wrappers for anytime-constrained MDPs. Includes a comprehensive benchmarking suite for evaluating performance tradeoffs. (*AISTATS '24*)
- Python **Knapsack Approximations Suite**, [\[GitHub Link\]](#)
 A benchmarking and evaluation framework for approximation algorithms, comparing classical polynomial-time methods against learning-based heuristics on large-scale resource allocation problems.

- Python **Strategic Manipulation in Multi-Agent Decision-Making**, *[GitHub Link]*
Reference implementations of polynomial-time algorithms for online attacks and defenses for Markov games, enabling empirical analysis of vulnerability and enhancing robustness of multi-agent decision-making systems. Demonstrated on MiniGrid environments. (AAAI '24)

Selected Talks

From Knapsacks to Self Driving: FPTAS Recipes for Constrained Reinforcement Learning

- UMD College Park (MARL Seminar) *March 2025*
- UW-Madison (Theory of Computing Seminar) *November 2024*

Optimal Attack and Defense for Reinforcement Learning

- UW-Madison (Theory of Computing Seminar) *February 2025*
- UW-Madison (Computer Vision Roundtable) *January 2024*

Teaching Experience

Primary Instructor

- Summer 2022 **CS 577: Introduction to Algorithms**, *University of Wisconsin-Madison*
- Responsible for curriculum design, lecture delivery, course structure, and student evaluation for a class of 100 students.
 - Designed original lecture slides, lecture notes, lecture videos, and problem sets focusing on the design of efficient algorithms and rigorous proofs of correctness.
 - Managed one TA and grader.

Head Teaching Assistant

- Spring 2021, 2022 **CS 540: Introduction to Artificial Intelligence**, *University of Wisconsin-Madison*
- Coordinated the administrative and grading workflow for a large-scale course with 500 students, 8 TAs, and over 20 graders and peer mentors.
 - Developed standardized grading rubrics and course structures in addition to the standard teaching assistant duties.
 - Instructors: Ilias Diakonikolas, Sharon Li, Fred Sala, and Jerry Zhu

Guest Lecturer

- Spring 2025 **CS 839: Game Theory, Optimization & Learning**, *University of Wisconsin-Madison*
- Designed and presented two lectures on the theory of multi-agent reinforcement learning.
 - Instructor: Manolis Vlatakis

- Summer 2021, 2022 **New Horizons in Theoretical Computer Science Summer School, ACM SIGACT**
- Designed lectures spanning complexity theory basics, interactive proofs, and Markov-chain Monte Carlo methods.
 - Instructors: Antonio Blanca, Nicole Immorlica, Yael Kalai, Jelani Nelson, and Ashia Wilson

Graduate Teaching Assistant

- Fall 2020, 2021 **CS 540: Introduction to Artificial Intelligence, University of Wisconsin-Madison**
Instructors: Anthony Gitter, Josiah Hanna, Yin Li, Yingyu Liang, and Daifeng Wang
- Spring 2019, 2020 **CS 577: Introduction to Algorithms, University of Wisconsin-Madison**
Instructors: Shuchi Chawla, Dieter van Melkebeek, and Christos Tzamos
- Fall 2019 **CS 301: Introduction to Data Science, University of Wisconsin-Madison**
Instructor: Tyler Caraza-Harter
- Fall 2018 **CS 524: Introduction to Optimization, University of Wisconsin-Madison**
Instructor: Steve Wright

Awards and Honors

- 2021–2025 **LEAP Fellow, CMD-IT / NSF Alliance**
Selected for the NSF-funded Leadership, Excellence, Access, and Pathways (LEAP) Alliance fellowship, focused on diversifying future faculty leadership in computing.
- 2023 **UW CS Graduate Instructor Award, University of Wisconsin-Madison**
Recognition for excellent performance as a primary instructor.
- 2019–2022 **The "Golden Brick" Teaching Award, University of Wisconsin-Madison**
A student/instructor-voted honor for most effective teaching. Won consecutively for every year served as a TA (4+ years).
- 2018–2019 **CS Departmental Scholarship, University of Wisconsin-Madison**
Merit-based scholarship awarded to incoming PhD students.
- 2018 **Most Outstanding Major Award, University of Illinois Urbana-Champaign**
Awarded to the #1-ranked student in Mathematics and Computer Science.

Professional Service

- 2024–2025 **Conference Reviewer**
NeurIPS (2024, 2025), ICML (2024, 2025), AISTATS (2025), RLC (2024)

Leadership & Organization

2023–2025 **Co-Organizer**, *Machine Learning Lunch Meetings (MLLM) Seminar*,
University of Wisconsin-Madison
Organized weekly research seminars for the ML community, which included managing speaker invitations, scheduling, and logistics for internal and external presenters.

Outreach & Mentorship

2021–2022 **Planning Committee & Teaching Assistant**, *ACM SIGACT*,
New Horizons in Theoretical Computer Science Summer School
A week-long summer school aimed at inspiring undergraduates from under-represented backgrounds to pursue careers in TCS.

- Served on the planning committee.
- Mentored students during technical sessions and social events.
- Developed and presented lessons on selected TCS topics.

Technical Skills

Languages **Python, Julia, C/C++, Mathematica, SageMath, $\text{\LaTeX}$, Bash**

Libraries **Gymnasium, Gurobi, SciPy, PyTorch, NumPy, Pandas**